MTH622 - Vectors and Classical Mechanics (QUIZ NO.2)

Quiz Start Time: 12:20 PM, 17 July 2023

Question # 3 of 10 (Start time: 12:21:35 PM, 17 July 2023)

Total Marks: 1

If a body rotates about some external fixed point it is called revolution

Select the correct option

- spin
- revolution

Click to Save Answer & Move to Next Question

B

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 1 of 10 (Start time: 12:20:02 PM, 17 July 2023) **Total Marks: 1**

Let 16kgm^2 be the momentum of inertia of a system of particles about AB, and 4kg be the total mass of the system then the radius of gyration of the system AB would be _____m.

Select the correct option

Reload Math Equations

- 8
- 2
- 1
- 4

Click to Save Answer & Move to Next Question

B

sec(s)

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 2 of 10 (Start time: 12:20:57 PM, 17 July 2023) Total Marks: 1

If an object explodes, the center of mass will

Select the correct option

- come into rest
- continue its previous motion

Click to Save Answer & Move to Next Question



sec(s)

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 6 of 10 (Start time: 12:23:53 PM, 17 July 2023) Total Marks: 1

A solid metal sphere having the moment of inertia of 0.03kgm^2 is rotating with an angular speed of 13.8 rad/s. Its angular momentum would be _____kgm^{2}s^{-1}.

Select the correct option

Reload Math Equations

- 0.204
- 0.414
- 0.354
- 0.440

Click to Save Answer & Move to Next Question



sec(s)

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 5 of 10 (Start time: 12:23:12 PM, 17 July 2023) Total Marks: 1

Let $56kgms^{-1}$ be the linear momentum of a system of particles and $4ms^{-1}$ be the velocity of the center of mass then the total mass of the system would be _____kg.

Select the correct option

Reload Math Equations

- 14
- 48
- 68
- 120

Click to Save Answer & Move to Next Question

A

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 4 of 10 (Start time: 12:22:10 PM, 17 July 2023) Total Marks: 1

Let 2kg be the mass and 0.5m be the radius of a right circular cone. Its moment of inertia about the axis of symmetry will be ____kgm^2.

Select the correct option

Reload Math Equations

- 0.50
- 0.42
- 0.35
- 0.15

Click to Save Answer & Move to Next Question



Question # 7 of 10 (Start time: 12:24:29 PM, 17 July 2023) **Total Marks:** 1

If no external force is acting on the system of two colliding bodies, then by the principle of conservation of linear momentum which of the following statement is correct?

Select the correct option

Reload Math Equations

○	Total momentum before the collision is less than the total momentum after the collision.
○	Total momentum before the collision is reciprocal of the total momentum after the collision.
○	Total momentum before the collision is greater than the total momentum after the collision.
○	Total momentum before the collision is equal to the total momentum after the collision.

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 8 of 10 (Start time: 12:25:05 PM, 17 July 2023) **Total Marks: 1**

If the total external torque on the system is zero, then its angular momentum will be _____.

Select the correct option

Reload Math Equations

- constant
- zero
- increasing
- decresing

Click to Save Answer & Move to Next Question



Time Left sec(s)

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 9 of 10 (Start time: 12:25:51 PM, 17 July 2023) Total Marks: 1

Let $\vec{r} = 2\hat{i}$ be the position vector of a particle and $\vec{p} = 3\hat{i}$ be its momentum then the magnitude of angular momentum $\vec{L}$ of the particle will be - - - - -

Select the correct option

Reload Math Equations

- 6
- 5
- 1
- 0

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz start time: 12:30 PM, 17 July 2023

Question # 3 of 10 (start time: 12:32:09 PM, 17 July 2023) Total Marks:

A solid metal sphere having the moment of inertia of $0.03 kgm^2$ is rotating with an angular speed of 13.8 rad/s. Its angular momentum would be _____ kgm^2s^{-1}.

Select the correct option

Reload Math Equations

- 0.354
- 0.304
- 0.414
- 0.440

Click to Save Answer & Move to Next Question

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:30 PM, 17 July 2023

Question # 1 of 10 (start time: 12:30:31 PM, 17 July 2023) Total Marks: 1

Let $64kgms^{-2}$ be the rate of linear momentum of a particle moving with the acceleration $16ms^{-2}$ then mass of the particle would be _____kg

Select the correct option

Reload Math Equations

- 14
- 8
- 28
- 4

Click to Save Answer & Move to Next Question

D

Time Left sec(s)

MTH522 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:20 PM, 17 July 2023

Question # 10 of 10 (Start time: 12:26:37 PM, 17 July 2023) Total Marks: 1

Let P be any point on the ellipsoid of inertia and 0.25kgm^2 be the moment of inertia about any line OP of the rigid body. The length of line OP will be _____m.

Select the correct option

Reload Math Equations

- 4
- 2
- 5
- 1

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:30 PM, 17 July 2023

Question # 6 of 10 (Start time: 12:34:52 PM, 17 July 2023) Total Marks:

If a system is isolated, then the total external torque (moment of force) remains constant in _____

Select the correct option

Reload Math Equations

- number of particles of the system
- direction
- magnitude
- both magnitude and direction

Click to Save Answer & Move to Next Question



Question # 7 of 10 (Start time: 12:35:55 PM, 17 July 2023) Total Marks: 1

Let $56kgms^{-1}$ be the linear momentum of a system of particles and $4ms^{-1}$ be the velocity of the center of mass then the total mass of the system would be _____kg.

Select the correct option

Reload Math Equations

- 120
- 68
- 46
- 14

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2)

Quiz Start Time: 12:30 PM, 17 July 2023

Question # 8 of 10 (Start time: 12:36:29 PM, 17 July 2023)

Total Marks: 1

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

Reload Math Equations

- perpendicular
- coincident

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2)

Quiz Start Time: 12:42 PM, 17 July 2023

Question # 6 of 10 (Start time: 12:47:44 PM, 17 July 2023)

Total Marks: 1

A solid metal sphere having the moment of inertia of 0.44kgm^2 is rotating with an angular speed of 2 rad/s.Its rotational kinetic energy would be _____ J.

Select the correct option

Reload Math Equations

- 0.48
- 0.76
- 0.88
- 0.66

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:42 PM, 17 July 2023

Question # 2 of 10 (Start time: 12:44:25 PM, 17 July 2023) Total Marks: 1

What happens to angular velocity when the moment of inertia of an isolated system is doubled to keep the angular momentum constant?

Select the correct option

Reload Math Equations

- It is halved.
- It becomes zero.
- It doubles.
- Remains same.

Click to Save Answer & Move to Next Question



sec(s)

MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 12:30 PM, 17 July 2023

Question # 10 of 10 (Start time: 12:38:06 PM, 17 July 2023) Total Marks: 1

Let $0.01 kgm^2$ be the moment of inertia (about an axis passing through the fixed point) of a rigid body rotating with the angular velocity 20 rad/s. Its kinetic energy will be ____ J.

Select the correct option

Reload Math Equations

- 2
- 1
- 4
- 5

Click to Save Answer & Move to Next Question



MTH622 - Vectors and Classical Mechanics (QUIZ NO.2) Quiz Start Time: 01:11 PM, 17 July 2023

Question # 5 of 10 (Start time: 01:14:46 PM, 17 July 2023) Total Marks: 1

Let $64kgms^{-2}$ be the rate of linear momentum of a particle moving with the acceleration $16ms^{-2}$ then mass of the particle would be _____kg.

Select the correct option

Reload Math Equations

- 4
- 6
- 14
- 28

Click to Save Answer & Move to Next Question



Question # 3 of 10 (Start time: 12:58:58 PM, 17 July 2023) Total Marks: 1

In general when a body rotates about fixed point, the angular velocity and the angular momentum vector about the fixed point are in the same direction.

Select the correct option

Reload Math Equations

- True
- False

Click to Save Answer & Move to Next Question



MTH622: Quiz 2

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Question # 3 of 10

SOLVED BY: WAQAR HUSSAIN

If a particle of mass 4 moving with velocity $4\hat{i}$ collides another particle of mass m with velocity $3\hat{i}$. If after collision the velocity of first particle is $5\hat{i}$ and that of second particle is $2\hat{i}$. Find m

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- 3
- 4 (Module 132, Pg.57)
-
-

TALEEMI MARKAZ * Enter To Learn *

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Question # 5 of 10

SOLVED BY: WAQAR HUSSAIN

Angular momentum can be defined as

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- $L =$
- $L =$
- $L =$
- $L =$

Angular Momentum

The angular momentum of a single particle is defined as the cross product of linear momentum and position vector of concerned particle.

Mathematically

$$L = r \times p$$

$$L = r \times mv$$

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Question # 4 of 10

SOLVED BY: WAQAR HUSSAIN

If a body rotates about some external fixed point it is called revolution

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- spin
- revolu
-
-

If the axis of rotation passes through the center of mass of the rigid body then body is said to be spin or rotate upon itself. If a body rotates about some external fixed point is called revolution or orbital motion of the rigid body.

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Question # 1 of 10

SOLVED BY: WAQAR HUSSAIN

If the axis of rotation passes through the center of mass of the rigid body then it is called

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- revolution
- spin
-
-

If the axis of rotation passes through the center of mass of the rigid body then body is said to be spin or rotate upon itself.

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Question # 8 of 10 **SOLVED BY: WAQAR HUSSAIN**

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$.

Select the correct option **PRESENTED BY: TALEEMI MARKAZ**

- ○ $I =$
- ○ $I =$
- ○
- ○

As we know that:

$$K^2 = \frac{I}{M} \quad \ldots Eq.(1)$$

Given that:

$$K = \sqrt{\frac{2}{5}a}$$

By putting the value of k in Eq.(1), we get

$$\left(\sqrt{\frac{2}{5}}a\right)^2 = \frac{I}{M} \implies \frac{2}{5}a^2 = \frac{I}{M}$$

$$\frac{2}{5}Ma^2 = I$$

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Question # 8 of 10

SOLVED BY: WAQAR HUSSAIN

What is the moment of inertia I of a hemisphere of mass M about an axis through its center given the corresponding radius of gyration is $K = \sqrt{\frac{2}{5}}a$

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- ○ $I =$
- ○ $I =$
- ○
- ○

As we know that:

$$K^2 = \frac{I}{M} \quad \ldots Eq.(1)$$

Given that:

$$K = \sqrt{\frac{2}{5}a}$$

By putting the value of k in Eq.(1), we get

$$\left(\sqrt{\frac{2}{5}}a\right)^2 = \frac{I}{M} \implies \frac{2}{5}a^2 = \frac{I}{M}$$

$$\frac{2}{5}Ma^2 = I$$

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Question # 7 of 10

SOLVED BY: WAQAR HUSSAIN

The torque acting on a particle is proportional to the time rate of change in its angular acceleration.

PRESENTED BY: TALEEMI MARKAZ

Select the correct option

- false
- true

Theorem:

The torque acting on a particle equals the time rate of change in its angular momentum, i.e.,

$$\tau = \frac{d\Omega}{dt}$$

where $\Omega = r \times p$ is defined angular momentum of the system.